HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed high-performance C++11 applications for edge computing on **Linux** platforms, focusing on memory management and CPU optimization.
- Implemented comprehensive unit and integration tests using **CMake** and C++ testing frameworks to ensure code quality and reliability for vehicle applications.
- Collaborated within cross-functional teams in an **Agile** environment, participating in sprint planning, code reviews, and contributing to the delivery of production-ready software.
- Designed and implemented low-latency communication protocols for in-vehicle systems, ensuring efficient data exchange between ECUs.
- Wrote **Python** scripts to automate build processes and manage dependencies for C++ projects, reducing manual build times by approximately **40%**.
- Integrated modern build tools like **Bazel** for efficient dependency management and parallel compilation of complex C++ codebases.
- Participated in **CI/CD** workflows, managing Jenkins pipelines for automated building, testing, and deployment of vehicle applications to target hardware.

SKILLS

Languages – C++, Python, Java, SQL, R

Tools & Technologies – Linux, CMake, Bazel, Docker, Jenkins, Git, Jira, Confluence, Splunk, Dynatrace

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

Databases – PostgreSQL, MySQL, Oracle

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn

Data Visualization – Power BI, Tableau

Methodologies – Agile, Scrum

PROJECTS

Real-Time Vehicle Data Acquisition System (HCL Technologies)

- Developed a C++ application for real-time data acquisition from vehicle sensors, ensuring low-latency processing using **Linux** system calls.
- Implemented a **CMake** build system for the C++ project, enabling cross-platform compatibility and streamlined compilation on development and target environments.
- Integrated the data acquisition module with a **Python**-based data logging service, capturing approximately **500** data points per second for diagnostics.

Automated Build and Test Framework for Embedded C++ (HCL Technologies)

- Created a CI/CD pipeline using **Jenkins** to automate the build, test, and deployment of embedded C++ firmware for automotive components.
- Designed shell scripts to orchestrate builds using **CMake** and execute test suites, reducing manual deployment efforts by **35%**.
- Containerized the build environment using **Docker** to ensure consistent build outcomes across different development machines and CI servers.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)